Introduction to Dynamic Routing

Task 1: List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

Static Routing	Dynamic Routing
IP address are manually typed in by the administrator using the ip route command.	Automatically assigned by DHCP routers exchanging updates through a routing protocol (e.g., RIP. EIURP).
S.R does not adjust automatically to network changes, admin must edit the configuration.	D.R automatically recalculates and updates routes when the topology changes.
NO CPU/Bandwidth are used as no updates are exchanged.	Uses some CPU and bandwidth to send and process periodic or triggered updates.
It is practical only for small, simple networks.	It is used for large networks with many routers and changing paths.
It has high Administrative effort — every route on every router must be maintained by hand.	It has low Administrative effort — routers maintain their own tables once the protocol is configured.

Static Routing	Dynamic Routing	Real-world analogy
The path is fixed by the admin and never changes on its own.	The path is worked out automatically by the routers.	Static routing is like giving someone one fixed set of directions and telling them to always use that exact route, no matter what. Dynamic routing is like using Google Maps, which automatically picks the best route for you every time.
If the route (link) is blocked, it keeps trying the same broken path until a human fixes it.	If the route (link) is blocked, a new path is found automatically.	If there's a traffic jam or accident, Google Maps automatically re-routes you — just like RIP automatically finds a new path when a link goes down.
Every new location needs a manually added route on every router.	New routers/networks are learned automatically once they join and start exchanging updates.	Static routing is like Swiggy having to manually tell every delivery partner the exact route to every new restaurant. Dynamic routing is like the Swiggy app automatically updating everyone's route the moment a new restaurant or delivery hub is added.

Task 2: Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

The pre-configured topology has three routers connected in series — Router0, Router1, and Router2 — each running RIP and each with its own LAN, connected through switches to a PC:

- Router0: LAN GigabitEthernet0/0 = 192.168.10.1/24 (PC0 = 192.168.10.2), WAN Serial0/0/0 = 10.0.0.1 to Router1
- Router1: WAN Serial0/0/0 = 10.0.0.2 to Router0, WAN Serial0/0/1 = 20.0.0.1 to Router2, LAN GigabitEthernet0/0 = 192.168.20.1/24 (PC1 = 192.168.20.2)
- Router2: WAN Serial0/0/0 = 20.0.0.2 to Router1, LAN GigabitEthernet0/0 = 192.168.30.1/24 (PC2 = 192.168.30.2)

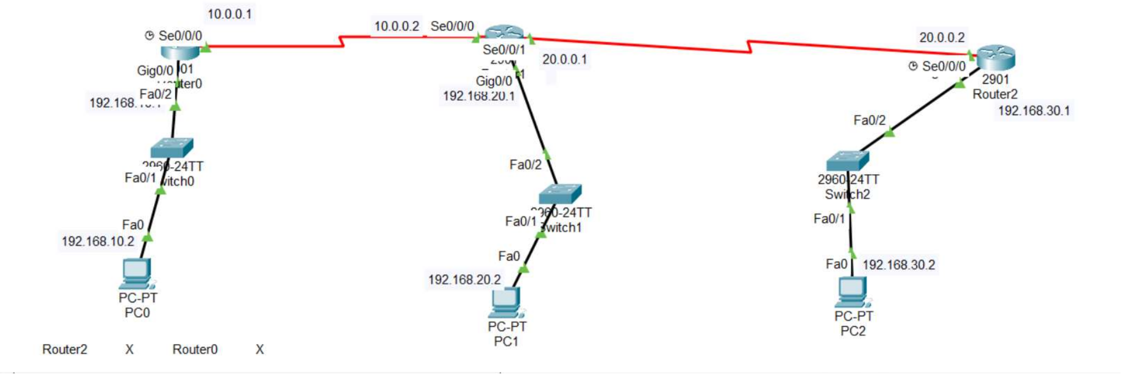


Figure 1: Three-router RIP topology — Router0 – Router1 – Router2

On Router0, show ip route shows the RIP-learned (R) routes to Router1's and Router2's networks:

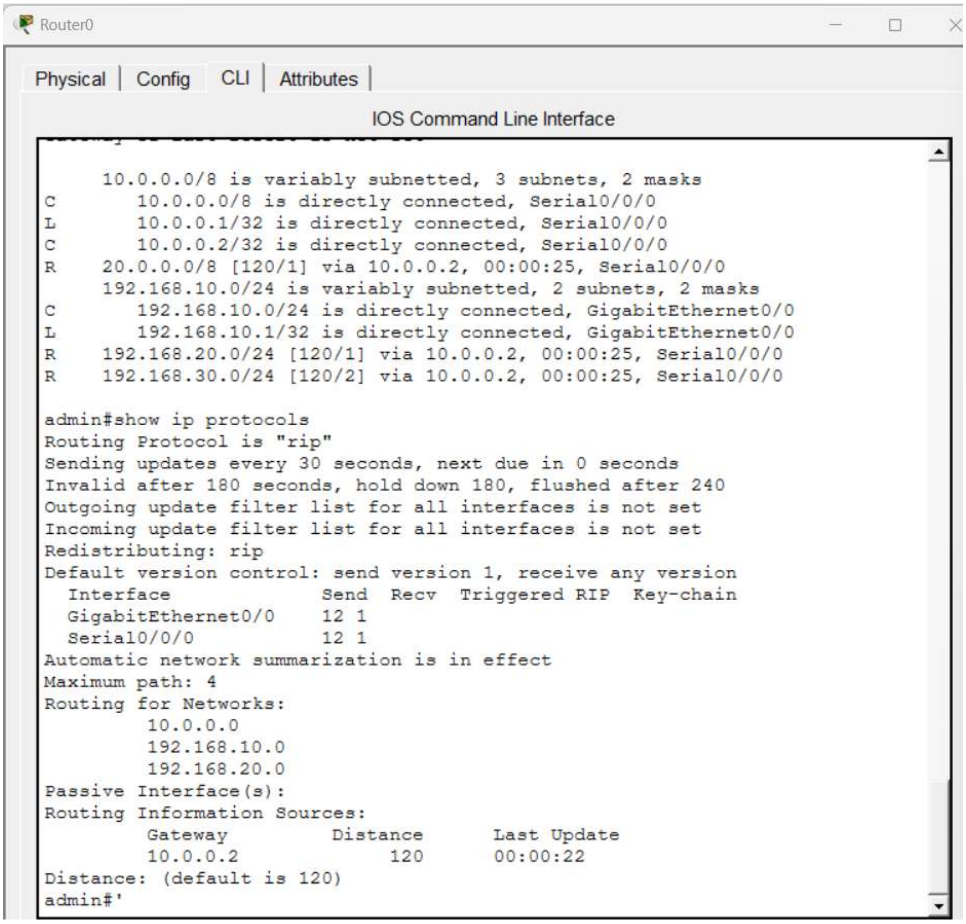
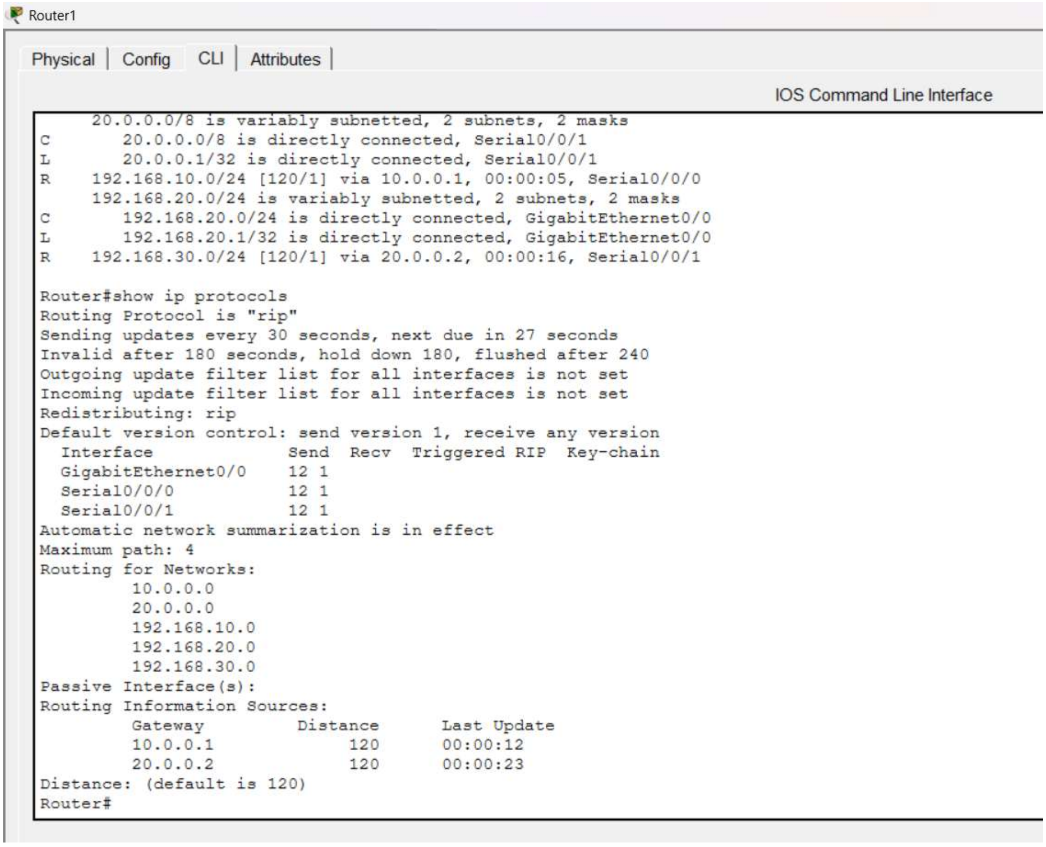


Figure 2: Router0 – show ip route and show ip protocols

Router0 shows 20.0.0.0/8 reachable at [120/1] via 10.0.0.2 (1 hop), 192.168.20.0/24 (Router1's LAN) also at [120/1], and 192.168.30.0/24 (Router2's LAN) at [120/2] — 2 hops, since traffic to Router2's LAN must pass through Router1 first. show ip protocols confirms RIP is running, sending updates every 30 seconds, with the usual timers (invalid after 180s, hold-down 180s, flushed after 240s) and an administrative distance of 120, learning routes from neighbour 10.0.0.2 (Router1).

On Router1 (the middle router), show ip route and show ip protocols show routes learned from both neighbours:



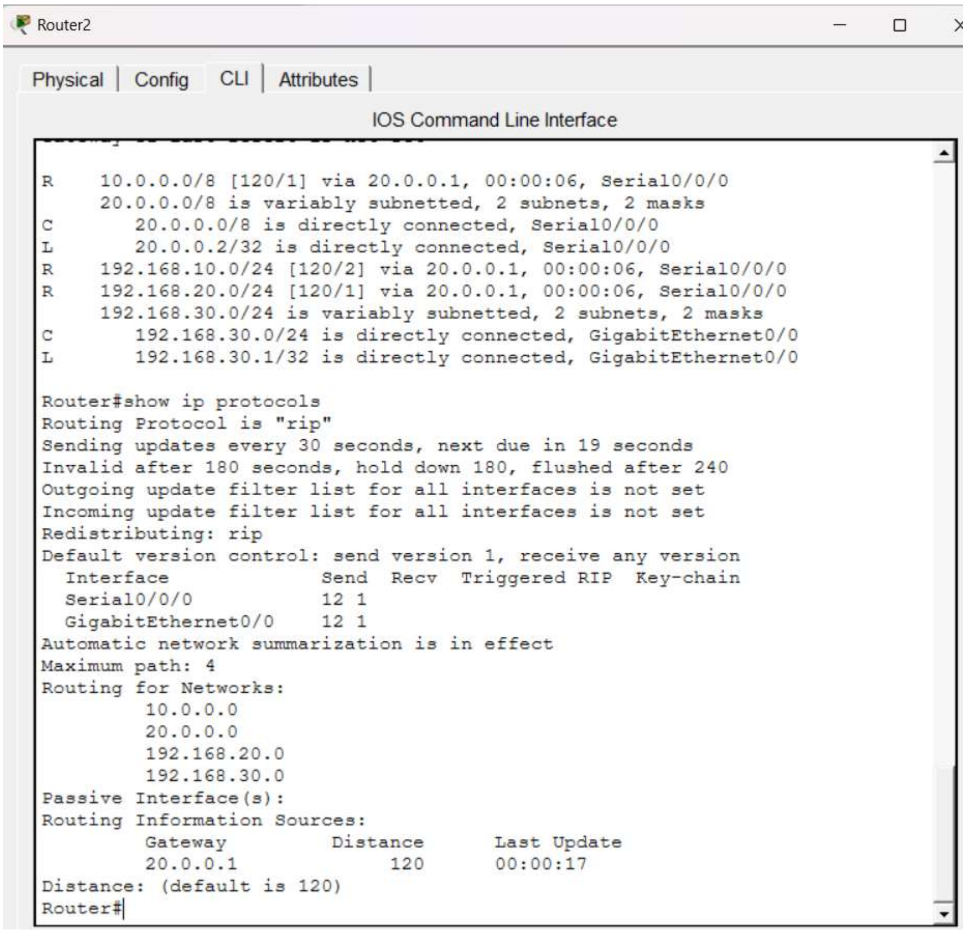

```
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Serial0/0/1
L       20.0.0.1/32 is directly connected, Serial0/0/1
R    192.168.10.0/24 [120/1] via 10.0.0.1, 00:00:05, Serial0/0/0
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0
R    192.168.30.0/24 [120/1] via 20.0.0.2, 00:00:16, Serial0/0/1

Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 27 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0    12 1
  Serial0/0/0           12 1
  Serial0/0/1           12 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
        10.0.0.0
        20.0.0.0
        192.168.10.0
        192.168.20.0
        192.168.30.0
Passive Interface(s):
Routing Information Sources:
        Gateway         Distance      Last Update
        10.0.0.1             120      00:00:12
        20.0.0.2             120      00:00:23
Distance: (default is 120)
Router#
```

Figure 3: Router1 – show ip route and show ip protocols

Router1 learns 192.168.10.0/24 (Router0's LAN) at [120/1] via 10.0.0.1, and 192.168.30.0/24 (Router2's LAN) at [120/1] via 20.0.0.2 — both just 1 hop away, since Router1 is directly connected to both neighbours. show ip protocols lists two routing information sources (10.0.0.1 and 20.0.0.2), confirming Router1 is exchanging RIP updates with both Router0 and Router2.

On Router2, show ip route and show ip protocols confirm the mirror image of Router0's view:




```
R    10.0.0.0/8 [120/1] via 20.0.0.1, 00:00:06, Serial0/0/0
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Serial0/0/0
L       20.0.0.2/32 is directly connected, Serial0/0/0
R    192.168.10.0/24 [120/2] via 20.0.0.1, 00:00:06, Serial0/0/0
R    192.168.20.0/24 [120/1] via 20.0.0.1, 00:00:06, Serial0/0/0
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0

Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 19 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface             Send  Recv  Triggered RIP  Key-chain
  Serial0/0/0           12 1
  GigabitEthernet0/0    12 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
         10.0.0.0
         20.0.0.0
         192.168.20.0
         192.168.30.0
Passive Interface(s):
Routing Information Sources:
         Gateway         Distance      Last Update
         20.0.0.1             120      00:00:17
Distance: (default is 120)
Router#
```

Figure 4: Router2 – show ip route and show ip protocols

Router2 shows 10.0.0.0/8 at [120/1] via 20.0.0.1 (1 hop to Router0's WAN link), 192.168.10.0/24 (Router0's LAN) at [120/2] — 2 hops away, and 192.168.20.0/24 (Router1's LAN) at [120/1] — 1 hop away. This matches Router0's view of Router2's LAN, confirming both ends of the topology agree on hop counts, and that RIP has fully converged across all three routers.

End-to-end connectivity was then verified from PC0 by pinging across the fully converged RIP network:

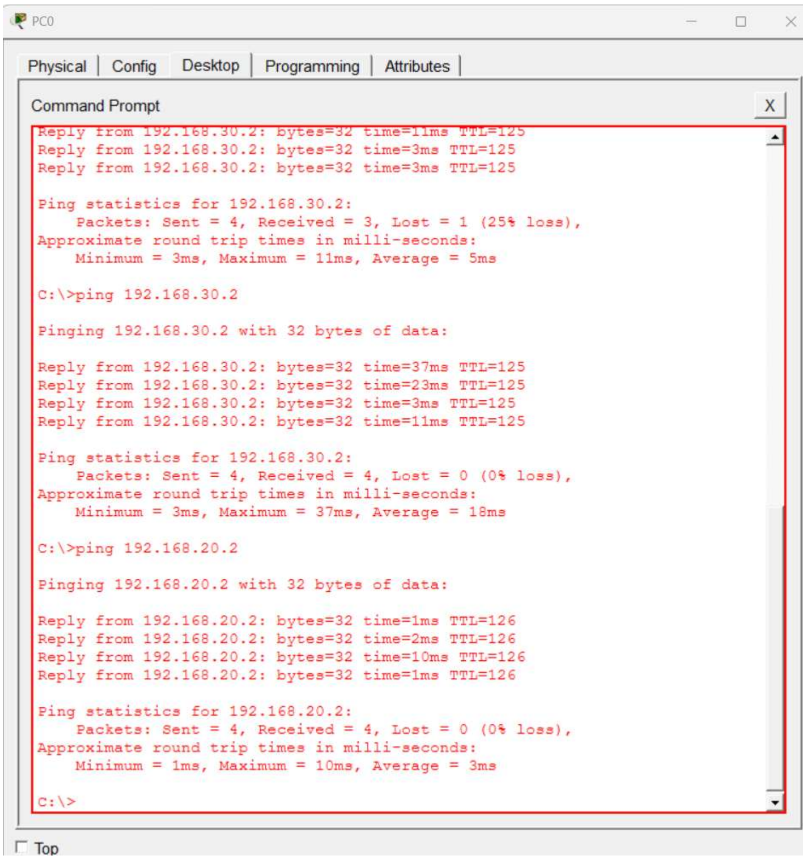


Figure 5: Ping tests from PC0 to 192.168.30.2 (Router2's LAN) and 192.168.20.2 (Router1's LAN)

The first ping from PC0 to 192.168.30.2 showed 1 timeout out of 4 packets (25% loss), most likely due to the router needing to resolve an ARP entry on the first attempt; repeating the same ping immediately after succeeded with all 4 packets received (0% loss). A ping from PC0 to 192.168.20.2 (Router1's LAN) also succeeded with 0% loss. This confirms that the RIP-learned routes are not just present in the routing tables but are actually working for real traffic across all three LANs.

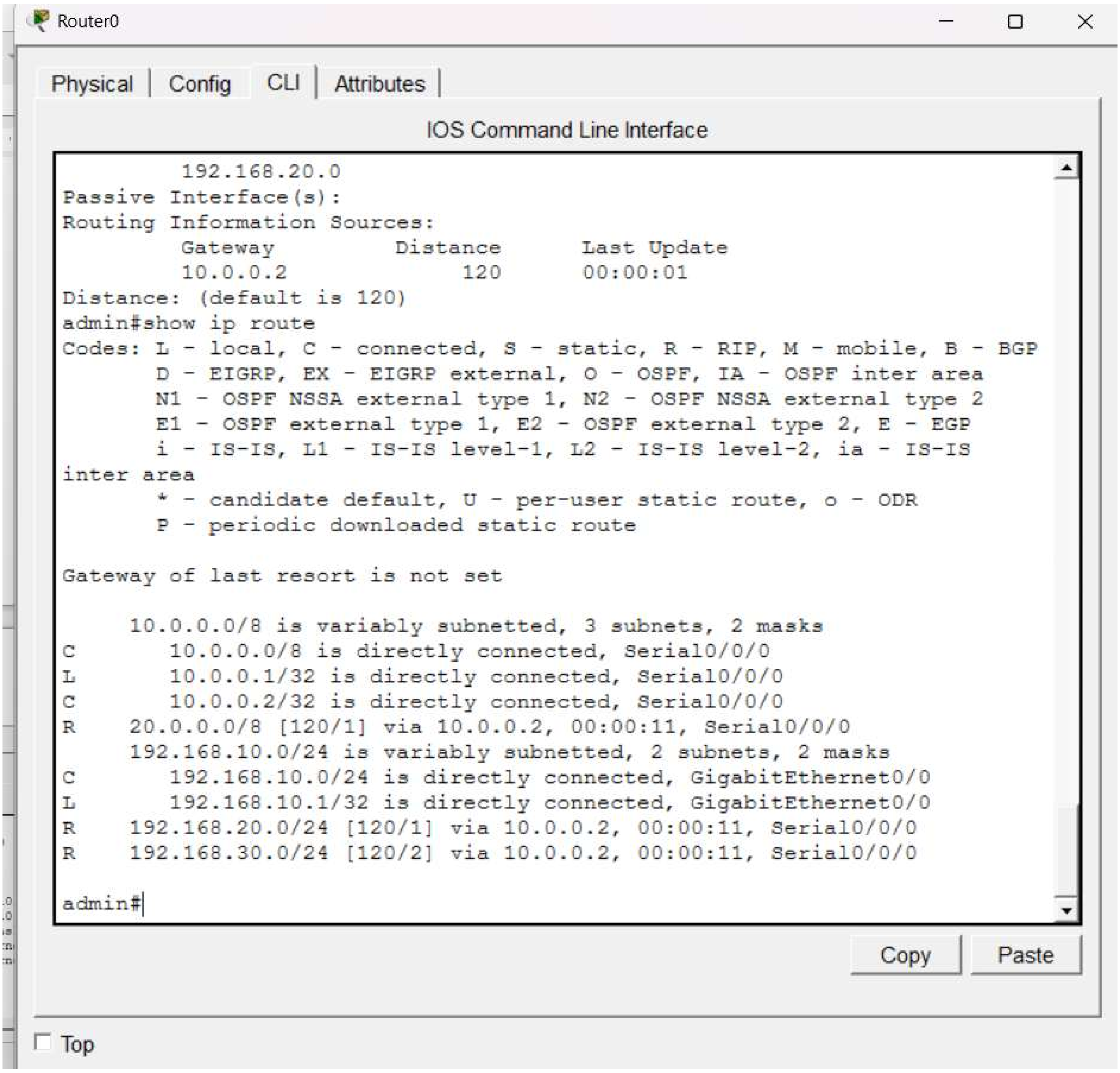
Task 3: Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route. Hint: Use the 'shutdown' command on an interface and monitor with 'show ip route' repeatedly.

To simulate a failure, the Serial0/0/1 interface on Router1 (its link to Router2) was shut down using the shutdown command in interface configuration mode:

```
Router1(config)# interface Serial0/0/1
Router1(config-if)# shutdown
```

Running show ip route repeatedly on Router0 afterward shows how RIP reacts in stages:

- Immediately: the route to 192.168.30.0/24 (Router2's LAN) may still briefly appear, since Router0 has not yet been told the link is down.
- Within a few seconds: because Router1 detects the interface going down instantly, it sends a triggered update marking the route as unreachable , which propagates the change quickly instead of waiting for the full 30-second update cycle.


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
192.168.20.0
Passive Interface(s):
Routing Information Sources:
Gateway Distance Last Update
10.0.0.2 120 00:00:01
Distance: (default is 120)
admin#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C 10.0.0.0/8 is directly connected, Serial0/0/0
L 10.0.0.1/32 is directly connected, Serial0/0/0
C 10.0.0.2/32 is directly connected, Serial0/0/0
R 20.0.0.0/8 [120/1] via 10.0.0.2, 00:00:11, Serial0/0/0
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
R 192.168.20.0/24 [120/1] via 10.0.0.2, 00:00:11, Serial0/0/0
R 192.168.30.0/24 [120/2] via 10.0.0.2, 00:00:11, Serial0/0/0
admin#
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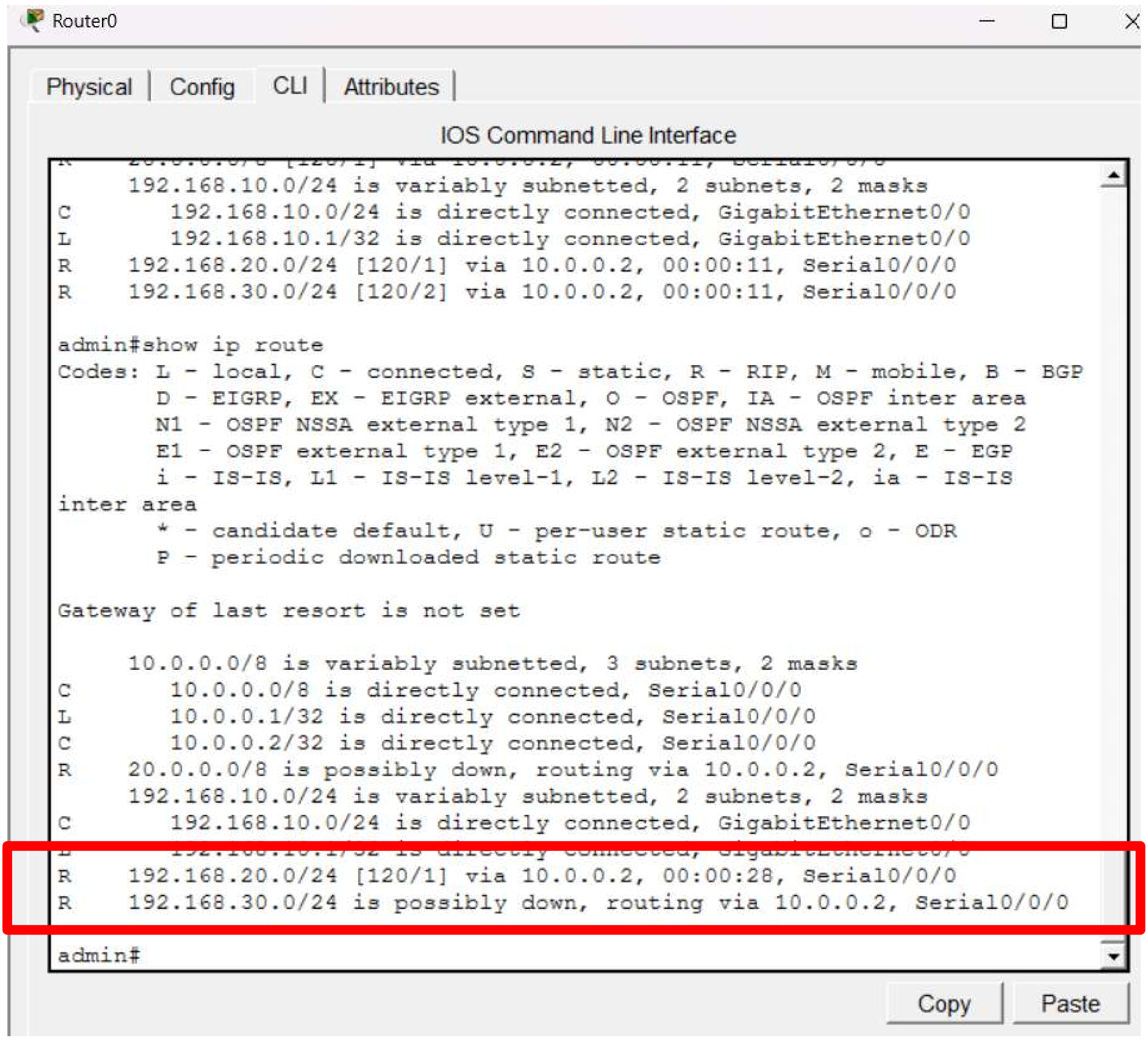
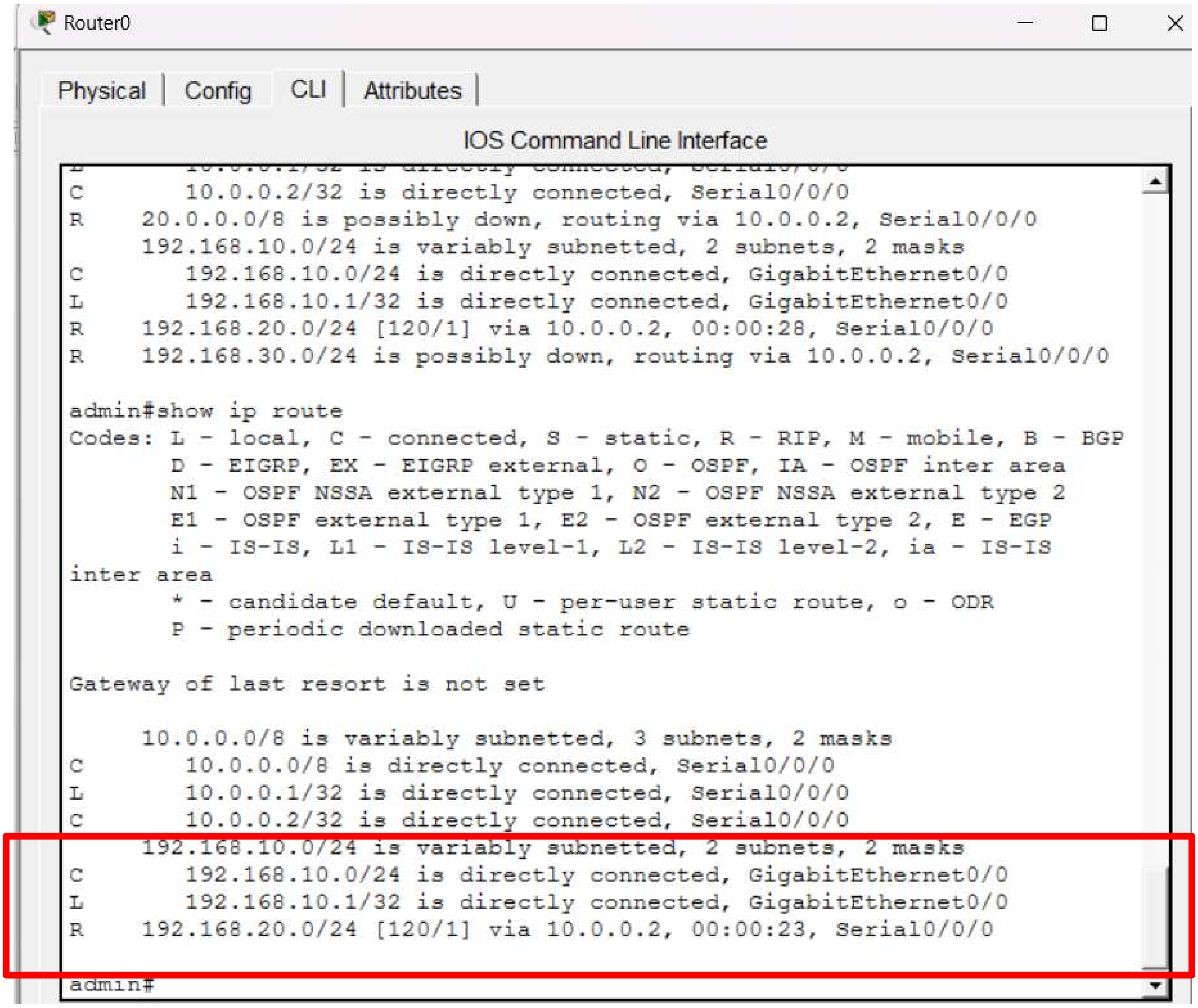


- Up to 180 seconds (invalid timer): if no triggered update were received, Router0 would mark the route as "possibly down" after not hearing about it for 180 seconds.



- Up to 240 seconds (flush timer): the route would be completely removed from the routing table 240 seconds after it was last heard, if it was never refreshed.

Task 4: Explain in 3-4 sentences how dynamic routing in apps like Uber or Zomato delivery tracking is similar to RIP dynamic routing in networks. Focus on how routes or paths are updated automatically in both cases.

Dynamic routing protocols like RIP and delivery-tracking systems in apps like Uber or Zomato both work by constantly sharing fresh information and automatically recalculating the best path, instead of relying on one route that is set once and never changes.

Just as RIP routers exchange updates every 30 seconds (or immediately when something changes) so every router always has current information, Uber and Zomato track a driver's live GPS location and instantly recompute the best route if there's traffic, a road closure, or the pickup point changes. In both cases, if part of the "network" becomes unreachable — a router link going down, or a road being blocked — the system automatically finds and switches to the next-best available path without needing a human to manually re-enter directions. This automatic, continuously-updated path selection is exactly what makes both RIP and app-based delivery tracking far more adaptable than a fixed, static route.